

Dozens of historical ciphers solved with LLMs

Large language models (Claude Opus 5, Opus 5.5, and Fable 5.1) solved more than forty historical ciphers in September 2026, working autonomously in an agentic setup with prompts designed by Dan Bourdeau (dnbourdeau@gmail.com), a data scientist and AI deployment engineer. Some of those previously unsolved ciphers had been published as public challenges. LLM agents found the others in databases and archives, with other agents handling research and cryptanalysis.

The LLMs recovered 24 keys using ciphertext-only cryptanalytic techniques. The remainder were solved using autonomous surveys of archives and other sources, looking for clues, possible plaintext, or keys published elsewhere.

Summaries of the solutions are listed below, including the primary source (the original document), the secondary source (where the item was found: DECODE, Tomokiyo's cryptiana, Cipherbrain, a library catalog), and a link to the full write-up with transcription, key, and reading. Cases where the key or a decipherment sat next to the ciphertext in the same file are not listed.

Full details and attempts on other ciphers are documented in <https://dbourdeau.github.io/cyphersolver/>

The solutions have been reviewed by Dr. George Lasry (george.lasry@gmail.com), an expert on cracking historical ciphers, who was part of a 2022-23 [project](#) to decipher dozens of encoded letters from Mary Queen of Scots found in the French National Library, and has published more than 30 [articles](#) and a [PhD thesis](#) on the subject.

This project is part of a recent wave of LLM solutions by other researchers (solving [Enigma messages](#), solving a [WWI German](#) cipher), following the introduction of powerful new frontier models in 2026. Lasry and Bourdeau are conducting a meta-analysis of the successes and limitations of current models against historical cipher challenges, based on extensive case studies from Bourdeau's project described here, and other projects. The research questions they are investigating include:

- Are LLMs effective in cracking historical ciphers because they tirelessly do extensive archival research, test hypotheses, look for patterns, and exhaustively look for cipher weaknesses, as humans could do but much faster, or because they are applying superior reasoning and are capable of designing novel sophisticated methods?
- While LLMs are not yet as effective as humans against some categories of cipher problems, is this only a question of time before newer models make human codebreaking redundant? Or are some of the limitations inherent and intrinsic and likely to leave some room (and fun) for human problem solving and codebreaking?

The ciphers are from the 15th to the 20th century. They include:

- **Substitution ciphers** where each letter of the alphabet is replaced by another letter or numerical code, or cipher symbols
- **Homophonic ciphers** where each letter of the alphabet can be encoded with more than one option, e.g., the relatively frequent letter e being replaced by either 20 or 21.
- **Polyphonic cipher**, in which every cipher symbol (or numerical code) may represent more than one plaintext letter, e.g., both e and f being encoded as 9.
- **Transposition ciphers**, in which the positions of letters of the plaintext messages are changed according to a pre-agreed transposition scheme.
- **Nomenclators and codebooks** - with one entry for each word or proper name.

1. Keys recovered from ciphertext only

In all those case studies, the key was rebuilt from the ciphertext alone, with no plaintext and no existing key available. Three items had also been solved earlier by George Lasry but never published; the LLMs solved them independently, and the LLM solutions match the solutions found by Lasry. For more details on the work of the LLM, its methods and steps, and the contents of the deciphered message, see the full write-up in the link provided below.

1. Four cipher letters to Gianfrancesco Gonzaga, lord of Mantua 1428-1430

To Gianfrancesco Gonzaga, lord of Mantua

“... dominus a[rchi]episcopus de Patras [...] cum licencia domini nostri dare volebat temporale dominium dominio Venetorum, et illud defenderet ab infidelibus. Dominus noster fecit proponere cardinalibus. Responderunt melius est perdere quam Venetis tradere, quia una vice recuperabitur; set si Veneti haberent, recuperare nunquam esset possibile ...”

Translation: “... the lord archbishop of Patras [...] wanted, with the leave of our lord [the Pope], to give the temporal lordship to the dominion of the Venetians, which would defend it against the infidels. Our lord had it put to the cardinals. They answered that it is better to lose it than to hand it to the Venetians, because one day it will be recovered; but if the Venetians had it, it could never be recovered ...”

Francesco de' Cattanei, Rome, February 13, 1430

Two keys rebuilt ciphertext-only: Rome, 9 Aug 1428 (96.6%) and Cattanei, Feb 1430 (96.7%). A third letter was re-deciphered from its interlinear gloss. Pandolfo Malatesta's own runs remain open. 91.5% of all signs read.

Method	ciphertext-only break · partial
Primary source	Archivio di Stato di Mantova, Archivio Gonzaga
Secondary source	DECODE R1854 (partially decrypted; catalogued as one letter)
Write-up	https://dbourdeau.github.io/cyphersolver/r1854.html

2. Sienese envoys and others to the government of Siena before 1429 to 1547

To the Signoria of Siena

“Noi aviamo chompr(e)so che questo signiore no chavalcharà per lo modo che chostoro di qui vorevero ... e farano la via di Laiaticho, e poi di là si partirano, e l'altro di saranno a Radichondoli ... E però richordiamo a la signioria vostra che ne detti luoghi faciate provvedere di pane, viada, vituagia ...”

Translation: “We have understood that this lord will not ride in the way those here would like ... and they will take the road by Lajatico, and then leave from there, and the next day they will be at Radicondoli ... And so we remind your lordship to have bread, fodder and victuals provided in the said places ...”

Sienese envoys (unnamed), undated

Keys recovered from the ciphertext alone for three letters (nos. 25, 18, 14) and from an alphabet fragment for a fourth; none of the 43 keys filed with the letters fits. No. 25 reads 96.2%, no. 18 (Latin treaty articles with the Emperor) 87.1%. Several other letters in the file remain open.

Method	ciphertext-only break · partial
Primary source	Archivio di Stato di Siena, Concistoro 2308, fasc. 2 "Lettere in cifra"
Secondary source	DECODE R4790-R4814 (no status); the archive's 1952 inventory calls the letters 'solo parzialmente decifrate'
Write-up	https://dbourdeau.github.io/cyphersolver/siena1421.html

3. Lope de Soria to Charles V: the key-A letters Genoa, June-July 1523

To Emperor Charles V

“... escriven de Milan como tenian nueva que era arrivato en Venecia un secretario del duque de Milan, el qual avia traído el mandado del Infante para concluir la liga con los venecianos, y que por esto el duque de Milan avia prometido al dicho Infante diez y ocho mil ducados, y que con esto tenian por cierto que se concluiria la dicha liga ...”

Translation: “... *they write from Milan that they had news that a secretary of the Duke of Milan had arrived in Venice, who had brought the Infante's mandate to conclude the league with the Venetians, and that for this the Duke of Milan had promised the said Infante eighteen thousand ducats, and that with this they held it certain that the said league would be concluded ...*”

Lope de Soria, Charles V's ambassador, Genoa, June 1523

Soria used two ciphers in 1523. Key A (the June letter in three copies and the body of the July letter) had no key and no crib; it was broken by simulated annealing on a Castilian language model of the 1520s and read nearly throughout: the Venetian league bought with Milanese money, the French king's expected descent on Italy.

Method	ciphertext-only break · partial · split entry: key A (June and July letters) only
Primary source	Real Academia de la Historia, Madrid, Colección Salazar y Castro A-28 (9/28)
Secondary source	DECODE R9488-R9498 (Non-decrypting)
Write-up	https://dbourdeau.github.io/cyphersolver/soria1523.html

4. Guido Rangone (Venice) to Anne de Montmorency 24 Dec 1529 and 10 Jan 1530

To Anne de Montmorency, Grand Master of France

“... 'l fa instantia a [...] de ven[i]re al servizio loro ... malissimo satisfato di lui; ha mandato a dirlo al [s]ignore Cesare e farli grande oferte ...”

Translation: “... *he urges [...] to come into their service ... very ill satisfied with him; he has sent to tell Signor Cesare so, and to make him great offers ...*”

Count Guido Rangone, condottiere in French service, Venice, January 10, 1530

A homophonic substitution of about 50 signs for Italian, broken by simulated annealing. Both passages concern Cesare Fregoso. 87.9% of signs read; four short runs are too short to resolve.

Method	ciphertext-only break · partial
Primary source	BnF, français 3070 f. 81 no. 40 and français 3082 f. 42 no. 19 · Gallica fr. 3082 · Gallica Clair. 331 copy
Secondary source	DECODE R4249 ; Tomokiyo, cryptiana (francis.htm) lists both as undeciphered · francis.htm
Write-up	https://dbourdeau.github.io/cyphersolver/rangone1530.html

5. Giovan Gioacchino da Passano (attributed), London, to Francis I 15 Mar 1530 (copy heading)

To Francis I, King of France

“... In denari contanti ... In baghe circa ... In vasele d'oro et d'argento ... In supectili de casa per el meno ... Per l'intrate del vescoato de Vinchestri et abadia de sancto Albano ...”

Translation: “... *In ready money ... In jewels, about ... In gold and silver plate ... In household furnishings, at least ... For the revenues of the bishopric of Winchester and the abbey of St Albans ...*”

Giovan Gioacchino da Passano (attributed), French agent,
London, March 15, 1530

An account of Wolsey's wealth: cash, credits, revenues, jewels and plate, Winchester, St Albans and York. A dotted re-transcription made the break possible. 386 of 402 signs read coherently

(96.0%).

Method	ciphertext-only break · complete
Primary source	BnF, Clairambault 331, ff. 156-157 · Gallica
Secondary source	Tomokiyo, cryptiana (francis.htm), listed as undeciphered · francis.htm
Write-up	https://dbourdeau.github.io/cyphersolver/joachim1530.html

6. A Bavarian agent to the Dukes of Bavaria: the Pressburg talks c. 1533-34

To the Dukes of Bavaria

“... von beden kunigen zu entlicher handlung gen Pressburg gesetzt ... der tag zu Pressburg erfolgt ist ... mit grossem pomp ... dem turkischen kaiser ... tribut ... zu geben ...”

Translation: “... set by both kings for a final negotiation at Pressburg ... the diet at Pressburg has taken place ... with great pomp ... the Turkish emperor ... tribute ... to give ...”

An unnamed Bavarian agent, c. 1533-34

Split entry. 3,150 signs in a homophonic substitution, broken by constrained annealing against German of 1472-1609 on the first run; a faint gloss on the first lines agrees word for word. It reports the Pressburg negotiations between Ferdinand and John Zápolya, a Turkish embassy, and Alvise Gritti. Running German with local errors; person codes open.

Method	ciphertext-only break · partial · split entry: R9416 pp. 3-4 (f. 263) only
Primary source	Bayerisches Hauptstaatsarchiv, Kurbayern Äußeres Archiv 4591, f. 263
Secondary source	DECODE R9416
Write-up	https://dbourdeau.github.io/cyphersolver/kaa4591.html

7. Claude de La Guiche, bishop of Mirepoix (Rome), to Montmorency 22 Nov 1551

To Anne de Montmorency, Constable of France

“... lequel est party pour s'en retourner a Sienes avec Dom Diego ... Le conte [de] Sainte Fiore est mal satisfait, et que facilement se reduiroit au service du roy ...”

Translation: “... who has left to go back to Siena with Dom Diego ... The Count [of] Santa Fiora is discontented, and would readily come over to the King's service ...”

Claude de La Guiche, Bishop of Mirepoix, French ambassador,
Rome, November 22, 1551

A 406-sign passage with 22 invented signs, broken by permutation annealing. It covers the Count of Santa Fiora, 'mal satisfait' and ready to serve the king on conditions. Seure's Lisbon letters in the same file are left open.

Method	ciphertext-only break · partial
Primary source	BnF, français 3138 no. 22, f. 60v · Gallica
Secondary source	BnF catalogue (Gallica harvest)
Write-up	https://dbourdeau.github.io/cyphersolver/guiche1551.html

8. Mémoire en chiffre to the connétable de Montmorency 12 Dec 1560

To Anne de Montmorency, Constable of France

“... puis peu de jours [il] s'est faict une assemblée en la ville de Clairac, en Agenois, d'environ deux mil rebelles, [ou] fust arresté et conclut que si le roy envoyoit forces de pardeça, qu'ils se met[r]oient en deffence, disans qu'ils aiment mieulx mourir les armes en main que si le roy les faisoit executer par ung bourreau.”

Translation: “... a few days ago [there] was held in the town of Clairac, in the Agenais, an assembly of about two thousand rebels, [where] it was resolved and

agreed that if the King sent forces to these parts, they would stand on their defence, saying that they would rather die with weapons in hand than be put to death by a hangman on the King's orders."

An unnamed Catholic gentleman of the Agenais, December 12, 1560

Ten pages in a cursive script: a homophonic substitution with two-stroke units and four word codes, broken from the images alone. It reports on the Protestant rising in the Agenais a week after François II's death. 98.9% of 1,715 cipher words read.

Method	ciphertext-only break · complete
Primary source	BnF, français 3157 no. 67, ff. 152r-156v · Gallica · BnF catalogue
Secondary source	BnF catalog (Gallica harvest); no DECODE record
Independent solution	Solved independently by the LLMs , from the ciphertext alone and without knowledge of any earlier solution. Lasry also solved it in 2023 (improved in February 2026), but his solution has not been published.
Write-up	https://dbourdeau.github.io/cyphersolver/memoire1560.html

9. "L'Estat du Roy de Navarre et de son party en France" after 1580

"... avec le prince de Parme pour lui rendre les places ... contre luy un party de malcontents ... sortir de la une armee de reistres pour la jeter et espandre sur le pais de celui qui ..."

Translation: *"... with the Prince of Parma, to hand the places over to him ... against him a party of malcontents ... to bring an army of reiters out of there, to hurl it and spread it over the country of the one who ..."*

Anonymous memoir on the King of Navarre and his party in France, after 1580

A French memoir on Navarre's party in a homophonic alphabet of about 50 graphic signs. The letter m is written as a group of dots, which earlier passes took for punctuation. 95.1% read.

Method	ciphertext-only break · complete
Primary source	British Library, Harley MS 1582, ff. 263-264
Secondary source	DECODE R85Q5 (Non-decrypted, "unknown sender")
Write-up	https://dbourdeau.github.io/cyphersolver/harley1582r85Q5.html

10. Henry of Navarre to Ségur 1585-86

To Jacques de Ségur-Pardaillan, his envoy to the German princes

"... capituler et conclure promptement avec ceux que vous adviserez ... avec le duc Casimir ou par son consentement, et faire la levée la plus grande et la faire marcher ..."

Translation: *"... to agree terms and conclude promptly with whomever you think fit ... with Duke Casimir or with his consent, and to raise the largest levy and set it marching ..."*

Henry of Navarre, the future Henry IV of France, October 1585

Three figure-cipher letters to the envoy raising a German army. The key, an alphabetical syllabary, was rebuilt by a structured annealer and checked on matched controls. The letters cover Duke Casimir, Clervant's reiters, and the road through the Vivarais.

Method	ciphertext-only break · complete
Primary source	BnF, 500 de Colbert 401, ff. 233, 239, 288v · Gallica
Secondary source	Tomokiyo, cryptiana unsolved list (as letters of Henry III) · unsolved.htm · henryiii.htm
Independent solution	Solved independently by the LLMs , from the ciphertext alone and without knowledge of any earlier solution. In private communications, Lasry wrote that he had also solved it in 2024, but his solution has not been published.
Write-up	https://dbourdeau.github.io/cyphersolver/segur.html

11. Thomas Edmondes (at Henri IV's court) to Lord Treasurer Burghley Jan 1592 - Jul 1594

To William Cecil, Lord Burghley, Lord Treasurer of England

"I doe secretelie learne that Monsr Montmorancy hath made great instance, as a condition whereof he required the performance before he would come, to haue the yong prince of Condé deliuered into the handes of Madame d'Angolesme his sister in law to be vnder his custodie."

Thomas Edmondes, Queen Elizabeth's agent at the court of Henri IV,
Tours, March 3, 1592

Six letters from the King's camp with lines in invented signs: a simple substitution of English with homophones and person signs, broken from the clear context. Everything reads except four lines on f. 59.

Method	ciphertext-only break · partial
Primary source	British Library, Stowe MS 166, ff. 50-61, 109-112
Secondary source	DECODE R7770-R7775 ("Lord Threr? to unknown recipient")
Write-up	https://dbourdeau.github.io/cyphersolver/stowe166.html

12. Matheo de Segura to the Constable of Castile, Madrid, 8 Aug 1596

To the Constable of Castile

"Un consejero muy grave me ha dicho que han hecho aquí junta sobre la cometa que ha auido estos días, y dizen que el mundo se acabará o morirá el rey por setiembre. Él bien cerca dello paresce que anda. Creo en Dios."

Translation: "*A most weighty councillor has told me that they have held a meeting here about the comet that has appeared these days, and they say that the world will end or the king will die by September. He does seem to be very close to it. I believe in God.*"

Matheo de Segura, Madrid, August 8, 1596

Two short passages in a homophonic cipher; the key was rebuilt by annealing and resegmenting the image. 119 of 125 cipher tokens read (95.2%). The passages concern predictions of the world ending or the king dying by September.

Method	ciphertext-only break · complete
Primary source	BnF, espagnol 336 no. 99, ff. 196r-198r · Gallica
Secondary source	Tomokiyo, cryptiana (spanish3.htm, unsolved.htm): the last es. 336 item listed as undeciphered · spanish3.htm · unsolved.htm
Write-up	https://dbourdeau.github.io/cyphersolver/segura1596.html

13. Ferdinand of Bavaria, Elector of Cologne, to Duke Maximilian I of Bavaria Bonn, 22 Dec 1619

To Duke Maximilian I of Bavaria, his brother

"... den Catolischen ins gemein und ohne unterschied [wird] geringer Zeit übrig sein, sich [vor] des andern Theils Einfal, Gwalt und Betrangnus zu entweren ... Conservation unser ... catolischer Religion und Seelen Seligkeit ..."

Translation: "*... for the Catholics, in general and without distinction, little time [will] be left to defend themselves [against] the other side's invasion, violence*"

and oppression ... the preservation of our ... Catholic religion and the salvation of souls ...

Ferdinand of Bavaria, Elector and Archbishop of Cologne,
Bonn, December 22, 1619

Ten pages in a homophonic substitution on two-digit figures with a nomenclator, broken by annealing 2,482 letters; the key reads the sibling R9425 unchanged. The letter covers quartering and troops six weeks after Frederick V's coronation. 95.8% read.

Method	ciphertext-only break · complete
Primary source	Bayerisches Hauptstaatsarchiv, Kurbayern Äußeres Archiv 4591, ff. 281-286
Secondary source	DECODE R9426 (non-decrypting, no author or recipient)
Write-up	https://dbourdeau.github.io/cyphersolver/ferdinand1619.html

14. Warsaw letter to the Bishop of Olmütz (?) 24 Dec 1627

To Cardinal Franz von Dietrichstein, Bishop of Olmütz (?)

“Mi ricordo che, fra li altri honori che [...] mi fece in Nicolsburg, mi confido d'hauer dato intentione a questa Serenissima [...] de uoler conferire un canonicato d'Olmiz ad uno de li [...] suoi figli ... si degni darne subito a me benigna risposta.”

Translation: “I recall that, among the other honours [Your Lordship] did me at Nikolsburg, I am confident I gave this Most Serene [Queen] to understand that [you] would confer a canonry of Olmütz on one of [her] sons ... deign to give me at once a kind answer about it.”

An unnamed agent at the Polish court, Warsaw, December 24, 1627

A homophonic substitution whose alphabet is in plain order, with syllables, nulls and thirteen word codes, recovered by annealing. The writer presses for a canonry of Olmütz for a son of the Queen of Poland. Read in full; the word codes are glossed from context.

Method	ciphertext-only break · complete
Primary source	ÖStA HHStA, Staatskanzlei Interiora, Chiffrenschlüssel Kt. 14 Fasc. 20 f. 176
Secondary source	DECODE R1408 (non-decrypting); transcription by Tomokiyo, cryptiana (variable2.htm, unsolved.htm) · variable2.htm · unsolved.htm
Write-up	https://dbourdeau.github.io/cyphersolver/warsaw.html

15. Sir Simonds D'Ewes, cipher log of his son's convulsion fits 3 Apr 1635 - Jan 1636

“... cum inter meos fuit amplexus alia terribilis convulsio ... Aug. xx. parum post lunae plenilunium (quae in diem xviii. istius mensis incidebat) tres difficillimas convulsiones, in quorum tertia os ipsius pene ad aurem dextram convulsum, quod ante numquam observavi.”

Translation: “... while he was in my arms, another terrible convulsion ... August 20, a little after the full moon (which fell on the 18th of this month), three very severe convulsions, in the third of which his mouth was drawn almost to his right ear, which I had never observed before.”

Sir Simonds D'Ewes, cipher log of his infant son's convulsion fits,
August 3 and 20, 1635

D'Ewes's private alphabet over Latin, rebuilt from cribs (convulsio, plena luna); the weekdays and moon phases check out against his Autobiography. 96.5% of 287 words read.

Method	ciphertext-only break · complete
Primary source	British Library, Harley MS 286, f. 61 · BL catalogue
Secondary source	DECODE R7759 (“unknown sender to unknown recipient, 1635”)
Write-up	https://dbourdeau.github.io/cyphersolver/harley286.html

16. Unknown writer to Hesse-Kassel undated (volume 1635-52)

“... das kann dieses orths halber observiren, dagegen den selben etwas tentiret werden solte ... dahero ich nicht unterlassen sollen, die notturft ... so wol allerhöchst ihrer Kays. Maiestet als auch ihrer Churf. Durchl. in Bayern ohnlängst nochmals ... anzudeuten ...”

Translation: “... *this can be watched from here, and something should be attempted against them ... I have therefore not failed to report the need ... both to His Most High Imperial Majesty and to His Electoral Highness in Bavaria, lately once more ...*”

An unnamed agent on the imperial side, undated (volume of 1635–52)

A digit-and-sign letter substitution. None of the volume's ~120 key sheets fits; the key was annealed on German of the period and confirmed by the contemporary gloss on lines 1-4. About 97% read.

Method	ciphertext-only break · complete
Primary source	Hessisches Staatsarchiv Marburg, HStAM 4 d Nr. 1218, no. 49 · Arcinsys
Secondary source	DECODE R4500 (partially decrypted)
Write-up	https://dbourdeau.github.io/cyphersolver/marburg1635.html

17. Ferenc Wesselényi, Palatine of Hungary, to Leopold I c. 1663-64

To Leopold I, Holy Roman Emperor and King of Hungary

“... presidium Maiestatis Vestre recipere. Quod si vero illud sponte et benevole acceptare nollent, extunc de ulterioribus faciendis actionibus temporis occasione nos accommodari debemus ... ista nux durissima tandem aliquando necessario frangi debet.”

Translation: “... *would receive Your Majesty's garrison. But if they would not accept it freely and willingly, then, as to the further actions to be taken, we shall have to suit ourselves to the opportunity of the moment ... this very hard nut must at last, of necessity, be cracked.*”

Ferenc Wesselényi, Palatine of Hungary, draft, c. 1663–64

A reversed-alphabet cipher with vowel homophones in a Latin draft, broken by annealing and then corrected by hand. A general is to take His Majesty's garrison by force if need be. 98.8% of cipher tokens read; two codes are open.

Method	ciphertext-only break · complete
Primary source	MNL OL E 199 (Wesselényi family archive), 8. pallium 1. 01
Secondary source	DECODE R674 (non-decrypted)
Write-up	https://dbourdeau.github.io/cyphersolver/wesselenyi1664.html

18. Potocka and Mniszech to Jakub Dunin 1714-16

To Jakub Dunin, her nephew

“... Pytałam się ieżeli ma co postanowionego. Ni ma nic. ... To tesz prawda że był u dominikanow wtenczas iak estancował z nią w polu ...”

Translation: “... *I asked whether he had anything arranged. He has nothing. ... It is also true that he was at the Dominicans' at the time when he was staying with her in the field ...*”

Barbara Potocka, née Dunin, undated (1714–16)

Potocka's alphabet was broken from a Polish postscript and carried over to eight French letters; 97.3% of Potocka tokens resolved. Mniszech's letter is still unread.

Method	ciphertext-only break · partial
Primary source	Archiwum Narodowe w Krakowie, Archiwum Sanguszków · szukajwarchiwach
Secondary source	DECODE R7524-R7530 , R7534-R7536
Write-up	https://dbourdeau.github.io/cyphersolver/potocka1714.html

19. Johann Heinrich Kauderbach (The Hague) to Brühl / Friedrich August II 1754-56

To Count Brühl, first minister of Saxony

“[On] a aujourd’hui des avis certains et positifs que le [traité] dont on parle depuis si longtemps a été signé à Versailles le deux de ce mois ... Cette nouvelle a causé une consternation extraordinaire ici parce qu’on craint qu’il n’y ait des articles secrets qui aillent encore plus loin.”

Translation: “[We] have today certain and positive reports that the [treaty] which has been talked of for so long was signed at Versailles on the second of this month ... This news has caused an extraordinary consternation here, because it is feared that there may be secret articles which go further still.”

Johann Heinrich Kauderbach, Saxon resident, The Hague, May 25, 1756

Unseparated figures in which 5 and 8 are nulls; one key serves all ten despatches. It was found by one-to-one annealing onto the vocabulary of his 1761 key. All ten despatches read at 65-97%, covering the Dutch augmentation debate and the Treaty of Versailles.

Method	ciphertext-only break · complete
Primary source	Koninklijk Huisarchief, The Hague, Prins Willem V inv. 201
Secondary source	DECODE R1043 , R1044 , R1954 , R1958 , R1959 , R2078-R2082 ("never solved" by the Hague cabinet)
Write-up	https://dbourdeau.github.io/cyphersolver/kauderbach1754.html

20. Gottfried van Swieten to Count Cobenzl Bonn, 1757-59

To Count Cobenzl, the Empress's minister at Brussels

“... subsister l’armée du roi y sans qu’il en coûte un sou ... il n’y a pas un officier général entendu à cette armée, excepté le duc de Broglie ...”

Translation: “... to keep the King's army there without it costing a sou ... there is not one capable general officer in this army, except the duc de Broglie ...”

Gottfried van Swieten, Austrian diplomat, Bonn, February 4, 1759

The 1759 table (groups 100-442) was rebuilt ciphertext-only by monotone annealing against French, then corrected by hand. The despatches cover the Gueldre convention and Soubise's army. 96% of 553 groups read.

Method	ciphertext-only break · complete
Primary source	Archives générales du Royaume, Brussels, Secrétairerie d’État et de Guerre 1236
Secondary source	DECODE R955-R957 (non-decrypte)
Write-up	https://dbourdeau.github.io/cyphersolver/swieten1757.html

21. An Orange prince to an officer of the Dutch émigré troops c. 1795

To an officer of the Dutch émigré troops in British pay

“... comme il pourroit arriver entretemps que le général Dundas vous déclarât qu’il cesseroit le payement, ou bien que les troupes devroient se laisser embarquer, il faudroit y souscrire et vous borner uniquement à tâcher d’obtenir [de] pouvoir donner une gratification à ceux qui ne voudroient se soumettre ...”

Translation: “... as it might happen in the meantime that General Dundas should declare to you that he would stop the pay, or else that the troops must let themselves be embarked, you would have to agree to it and confine yourself

solely to trying to obtain leave to give a gratuity to those who would not submit
 ...”

A son of William V, Prince of Orange, c. 1795

Vertical digit pairs from a 6x6 square, recovered by quadgram annealing on the Dutch postscript; the square then read the French page unchanged. The letter concerns General Dundas, pay and embarkation. 94% of cipher tokens read; about thirty word signs open.

Method	ciphertext-only break · partial
Primary source	Koninklijk Huisarchief, The Hague, Prins Willem V inv. 339
Secondary source	DECODE R1892 (non-decrypted)
Independent solution	Solved independently by the LLMs , from the ciphertext alone and without knowledge of any earlier solution. In private communications, Lasry wrote that he had also solved it in 2021, but his solution has not been published.
Write-up	https://dbourdeau.github.io/cyphersolver/r1892.html

22. Cardinal Soglia to nuncio Viale Prelà at Innsbruck Rome, 15 June 1848

To Mgr Michele Viale Prelà, papal nuncio at Innsbruck

“... per nuove sopraggiunte circostanze Le partecipo [...] del Padre che Ella rimanga in Inspruck presso l’Imperatore e ovunque egli fosse per recarsi; intendendosi sospesa la indicata delibera [...] relativa alla domanda de’ passaporti ... nuova istruzione, che Le viene indirizzata per tre diverse vie ...”

Translation: “... because of new circumstances that have arisen, I inform you [...] of the [Holy] Father that you are to remain at Innsbruck with the Emperor, and wherever he may go, the aforesaid decision [...] concerning the request for the passports being understood as suspended ... new instruction, which is being sent to you by three different routes ...”

Cardinal Giovanni Soglia, Secretary of State to Pope Pius IX, Rome, June 15, 1848

A 64-cell two-digit table with homophones and syllables, 5 as word separator, and an alphabetical one-part code. It is a counter-order: stay with the Emperor and suspend the request for passports. Ten code words read from context; 11 code groups open.

Method	ciphertext-only break · complete
Primary source	L'Italia del Popolo no. 40 (Milan, 30 June 1848), Roman broadsheet reprint
Secondary source	Cipherbrain challenge (Klaus Schmeh, 13 Dec 2014) · Cipherbrain, 13 Dec 2014
Note	also a Cipherbrain challenge (Klaus Schmeh)
Write-up	https://dbourdeau.github.io/cyphersolver/soglia1848.html

23. Swatow telegram to Sun Yat-sen 3 Apr 1916

To Sun Yat-sen's household in Tokyo (telegraphic address “Rosamonde”)

“潮城由莫擎宇獨立。我軍亦光復汕頭。後莫率大隊來，令我退出鎮守府。我軍力薄，暫由翼□支持。文慧返.....”

Translation: “Chaozhou city has been declared independent by Mo Qingyu. Our force too has recovered Swatow. Then Mo came with a large body of troops and ordered us out of the garrison headquarters. Our force is weak; for now we are supported by Yi-□. Wenhui is returning”

“Tanaka”, a Japanese with Sun Yat-sen's men, Swatow, April 3, 1916

Standard Chinese telegraph code passed through a systematic code condenser, recovered by brute force. 41 of about 44 characters read.

Method	ciphertext-only break · complete
Primary source	JACAR (Japan Center for Asian Historical Records) B03050738800 · JACAR
Secondary source	Tomokiyo, cryptiana list and blog (12 Dec 2021), unsolved since 2021 · chinesecrypto e.htm · blog, 12 Dec 2021
Write-up	https://dbourdeau.github.io/cyphersolver/sunyatsen.html

24. Sun Yat-sen's intercepted telegrams Mar 1916 - Mar 1917

To Sun Yat-sen's office in Tokyo

“英士昨下午在山田家被兇[轟?]斃，捕兇[一?]人，關係者數人。”

Translation: “Yingshi [Chen Qimei] was [shot?] dead by assassins yesterday afternoon in the Yamada house; [one?] assassin and several people involved arrested.”

Sun Yat-sen's party, Shanghai, May 19, 1916

About 60 condenser telegrams read under seven keys of the Swatow family, plus the Hankow +111 digit code, including Chen Qimei's assassination as reported to Tokyo. Sun's six-vowel 文密 code is still open.

Method	ciphertext-only break · complete
Primary source	JACAR, 特定人ノ発受郵便物取調 (Ministry of Communications intercepts) · JACAR (first file)
Secondary source	Follow-on from the Swatow telegram (cryptiana) · chineseencrypt e.htm
Write-up	https://dbourdeau.github.io/cyphersolver/sunintercepts.html

2. Keys rebuilt from a plaintext found outside the collection

*The key was rebuilt by LLMs, by aligning the ciphertext with a **plaintext** found elsewhere, with extensive, autonomous search in archives, book editions, archive calendars, websites, articles, or the decipherment of another document originally enciphered with the same key.*

25. Giuliano Caprile and Alfonso Cistarelli (Eger, Buda) to the Este 1519-21

To Alfonso I d'Este, Duke of Ferrara

“... il custode ... è tenuto Agria ... in castello ... deli danari del suo ... lui m'ha risposto dice de voler andar in Italia ...”

Translation: “... the keeper ... Eger is held ... in the castle ... of his money ... he has answered me, saying he wants to go to Italy ...”

Giuliano Caprile, canon of Eger, April 6, 1521

Somogyi's two plaintexts were used as a crib to rebuild the 1520-21 homophonic sign cipher by EM alignment. The key then read in part two letters with no decipherment anywhere (R1139, R1136): the custode and governor of Eger, the late bishop's money.

Method	key from external plaintext · partial
Primary source	Archivio di Stato di Modena, Ambasciatori, Ungheria b. 4 · inventory (PDF)
Secondary source	DECODE R1128-R1139 ; Vestigia · Vestigia
Key or plaintext from	W. Somogyi's printed decipherments of two letters, Scriptorium VI (2025) · REAL
Write-up	https://dbourdeau.github.io/cyphersolver/caprile1519.html

26. Adrian of Utrecht, the Admiral and the Constable of Castile to Charles V Vitoria, 30 Dec 1521

To Emperor Charles V

“... el fallecimiento del Rey de Portugal ... y agora con este moço el Rey que sucede en aquel Reino, y con que diz que tiene cabe sí algunos que no le dexan las orejas ociosas, no sabemos la demostración que querrá hacer; porque ... los Reyes de Portugal tienen alianza antigua con los de Francia ...”

Translation: “... the death of the King of Portugal ... and now, with this youth who succeeds as King in that realm, and who they say has some people about

him who do not leave his ears idle, we do not know what stance he will take; for ... the Kings of Portugal have an ancient alliance with those of France ...

Adrian of Utrecht, regent of Castile and future Pope Adrian VI, with the Admiral and the Constable of Castile, Vitoria, December 30, 1521

The ciphertext was transcribed from the images and aligned with the printed decipherment, rebuilding a mixed nomenclator. The rebuilt key read or corrected about 17 of the ~25 groups the print left blank, and showed that the dead king is Manuel I of Portugal, not the print's "Rey de Inglaterra".

Method	key from external plaintext · complete
Primary source	Archivo General de Simancas, Estado leg. 8 no. 150 · PARES
Secondary source	PARES ("cifrada prácticamente en su totalidad")
Key or plaintext from	Gredilla's 19th-century decipherment printed by Danvila, Memorial Histórico Español 38 (1899), pp. 703-705, found by OCR search of the archive.org scan · archive.org
Write-up	https://dbourdeau.github.io/cyphersolver/adrian1521.html

27. Hugo de Moncada to Charles V Monaco, 6 Oct 1524

To Emperor Charles V

“... duque de Borbon ... me rogaua que le lleuase a España, [...] diziendome muchas razones para ello. Yo le he respondido que vna vez ponga el exercito de V. Md. en seguro, y que estas galeras no tienen pan ...”

Translation: “... the Duke of Bourbon ... asked me to take him to Spain, [...] giving me many reasons for it. I have answered him that once he has put Your Majesty's army in safety, and that these galleys have no bread ...”

Hugo de Moncada, captain-general of the imperial galleys,
Monaco, October 6, 1524

442 cipher signs were transcribed and aligned with the print by EM, rebuilding a homophonic key with code groups. The key reads the print's gap: Bourbon asked to be carried to Spain in the galleys.

Method	key from external plaintext · complete
Primary source	Biblioteca Nacional de España, MSS/20213/12 · BNE Digital
Secondary source	DECODE R1191 (filed without author)
Key or plaintext from	The decipherment printed in Colección de documentos inéditos XXIV (1854), pp. 417-419, to which a pencil note on the leaf points · archive.org
Write-up	https://dbourdeau.github.io/cyphersolver/moncada1524.html

28. Burghley to Walsingham, letter-book copy early 1573

To Francis Walsingham, English ambassador in France

“... I see there was a great mistaking of our doings, for by Davison's writing of a ship sent to [St Valery], [...] imagined that Hall had a meaning to provoke Glasgow to come hither ... This you must think must needs appear very strange.”

William Cecil, Lord Burghley, Lord Treasurer, early 1573

A six-sign place name that recurs in four letters printed undeciphered in Digges's Compleat Ambassador (1655) reads VALERY. Only this place name is read (6 of 26 cipher tokens); the rest waits on a key not online.

Method	key from external plaintext · partial
Primary source	British Library, Harley MS 260 f. 419
Secondary source	DECODE R8364
Key or plaintext from	A French letter in Walsingham's file asking for a ship to be kept waiting at St Valery (BL Cotton Vespasian F VI), identified from the BL catalogue
Write-up	https://dbourdeau.github.io/cyphersolver/r8364.html

29. Sir Francis Walsingham to Edward Wotton, ambassador in Scotland Jul-Sep 1585

To Edward Wotton, English ambassador in Scotland

“Sir, 39 [Master of Gray] doth complain in a late letter written to his friend here, whom you know, of the alteration of your countenance towards him ... he offereth to make a most manifest proof, if 28 and 29 [the banished lords?] would come down to the Borders to join with 20 [Morton] and some other friends as he can procure, without the which he feareth that 10 [the King] will be carried away from 36 [England] ...”

Sir Francis Walsingham, Principal Secretary to Elizabeth I,
at the Court, August 21, 1585

Signed letters in clear English with a two-digit name code; the calendar's glosses fix 19 Arran, 39 Gray, 40 Justice Clerk, 24 Farnihurst and 20 Morton. Two uncalendared letters transcribed here. About fourteen sign-alphabet words on 10 Sept remain open.

Method	key from external plaintext · partial
Primary source	British Library, Add MS 32657 ff. 125-195
Secondary source	DECODE R4838-R4840 , R4844 (non-decrypte
Key or plaintext from	The bracketed glosses in the Calendar of State Papers, Scotland viii (1914), nos. 40, 41, 65, 106 · HathiTrust
Write-up	https://dbourdeau.github.io/cyphersolver/walsingham1585.html

30. Alchemical cipher note on a flyleaf (Speculum volume) undated (volume 1618)

“Prima materia ist die Galle vom Widder. Das Gefäß[β] ist eine halbrunde Kugel. Der Ofen ist die Waage.”

Translation: “*The prime matter is the gall of the ram. The vessel is a half-round sphere. The furnace is the balance.*”

Anonymous alchemical note on the flyleaf of the Rosicrucian Speculum sophericum rhodo-stauroticum, undated (volume of 1618)

Rous's partial reading was used as a crib and the substitution completed from the original image: PRIMA MATERIA, VOM WIDDER, OFFEN. All 79 signs assigned; 78 coherent without correction.

Method	key from external plaintext · complete
Primary source	Zentralbibliothek Zürich, SCH R 201 · e-manuscripta
Secondary source	DECODE R4817 (non-decrypte
Key or plaintext from	Anne-Simone Rous's partial reading, published in U. M. Korn, Bilderfahrzeug (2023), p. 178 n. 41 · doi
Write-up	https://dbourdeau.github.io/cyphersolver/r4817.html

31. The Venetian ambassador at Rome to the Doge and Senate 29 Apr 1628

To the Doge and Senate of Venice

“... col Cardinal Barberino di parlargli di quel negotio che lei mi si è commesso con cotesto Eccelso Consiglio, et di cavar anche qualche intimo particolare desiderato da lei ... avendomi ricercato ... se Palma sarà ben fortificata, quanto presidio aveva, se il passo della Pontebba si era ben assicurato ...”

Translation: “... with Cardinal Barberino, to speak to him of that business which you entrusted to me with that Most Excellent Council, and also to draw out some secret particular that you desired ... having asked me ... whether Palma will be

well fortified, how large a garrison it had, whether the pass of the Pontebba had been well secured ...”

The Venetian ambassador, Rome, April 29, 1628

R1874's decipherment (~4,800 letters) was aligned with its cipher by EM to give a syllabary grid, which reads this Rome despatch unchanged. 1,280 groups transcribed; 99.1% have a value. Cardinal Barberini asks whether Palmanova is fortified and the Pontebba pass secured.

Method	key from external plaintext · complete
Primary source	Archivio di Stato di Venezia, IT ASVe 0045 010, Busta 27 f. 354
Secondary source	DECODE R1862 (Non-decrypting, no transcription)
Key or plaintext from	The contemporary decipherment of a different despatch in the same cipher, Mocenigo from Madrid (Busta 30, DECODE R1874)
Write-up	https://dbourdeau.github.io/cyphersolver/r1862.html

32. Fra Giovanni di Lucca to Ferdinand III 30 May 1644

To Emperor Ferdinand III

“Al principe di Transil[u]ania lo impedirò con ogn[i] modo che il Turco non li dia aiuto, e farò rivoltare contra di lui, se sarà bisogno, il principe di Bog[o]dania, et li darò dumilia Cosachi, et manderò a voi con una sua lettera all’Imperatore che non faccia pace fin che non l’habbi humiliato o vinto.”

Translation: “As to the Prince of Transylvania: I shall by every means prevent the Turk from giving him help, and if need be I shall make the Prince of Moldavia rise against him and give him two thousand Cossacks, and I shall send [him] to you with a letter of his to the Emperor, that he make no peace until I have humbled or defeated him.”

Fra Giovanni di Lucca, reporting the words of Grand Hetman Stanisław Koniecpolski, Brody, May 30, 1644

Eight crib letters fixed from Tomokiyo's opening seeded an annealer that found a 24-figure Italian alphabet (all seeds agreeing, shuffled controls failing). Read in full: Grand Hetman Koniecpolski offers to keep the Turk off Rákóczi and supply 2,000 Cossacks.

Method	key from external plaintext · complete
Primary source	ÖStA HHStA, Staatskanzlei Interiora, Chiffrenschlüssel Kt. 20 Fasc. 27 ff. 18-19
Secondary source	DECODE R2159 ; Tomokiyo, cryptiana unsolved list · unsolved.htm
Key or plaintext from	Tomokiyo's published partial decipherment of the opening, "Al principe di trenti..." (cryptiana variable2.htm) · variable2.htm
Write-up	https://dbourdeau.github.io/cyphersolver/lucca.html

33. Feuquières to Catinat Pignerol, 1691

To Nicolas de Catinat, commanding the French army in Piedmont

“... le peu de monde qu'il y a et les moyens d'y entrer, tout doi[ven]t faire croire qu'ils ne viendront pas à le défendre sérieusement. Je ne sçais du tout par où faire passer à Saint-Ambroise les quatre-vingts maîtres que vous [vous] demandez ...”

Translation: “... the few men there are and the ways of getting in, all must make one believe that they will not come to defend it seriously. I have no idea at all by what road to get to Saint-Ambroise the eighty troopers you ask [of us] ...”

Antoine de Pas, marquis de Feuquières, governor of Pignerol, January 25, 1691

The 418-group two-part "petit chiffre" of the Pignerol governors was rebuilt by aligning the Louvois letter with its translation and carrying the 72 shared groups over. 586 of 601 tokens read: Feuquières sets out the surprise of Veillane (Avigliana).

Method	key from external plaintext · complete
Primary source	Service historique de la Défense (dépôt de la guerre); printed in Mémoires et correspondance du maréchal de Catinat (1819), t. II pp. 283-284
Secondary source	Tomokiyo, cryptiana (louisxiv.htm) · louisxiv.htm

Key or plaintext from	Another letter in the same code, Louvois to d'Herleville of 6 Sept 1690, printed with its contemporary translation in the same <i>Mémoires</i> , t. I pp. 140-142 · Mémoires (BSB scan) · Mémoires (BSB scan)
Write-up	https://dbourdeau.github.io/cyphersolver/feuquieres.html

34. John Armstrong to James Madison, coded postscript 1808

To James Madison, Secretary of State

“Russel ought to be the consul: he is an American by birth, and is much better qualified than any other candidate. In a word, he is above men in general. Next to him in fitness is O'Mealy, but he is, like Warden, an Irish[man].”

John Armstrong, United States Minister to France, Bourbon-l'Archambault, August 30, 1808

A 580-group code table (the "THE = 972" code) was rebuilt from the clerk's decodes in another archive, with the remaining groups inferred from their alphabetical slot. It reads 48 of the postscript's 49 groups; the 49th is a probable slip for "man".

Method	key from external plaintext · complete
Primary source	Library of Congress, Madison Papers, reel 10 frame 0521 · LoC
Secondary source	Tomokiyo, cryptiana unsolved list; Founders Online (postscript undeciphered) · unsolved.htm
Key or plaintext from	A State Department clerk's pencil decodes over Armstrong's 1806-07 despatches (NARA microfilm M34, roll 13), merged with Tomokiyo's partial table from a known-plaintext letter · NARA · decode20.htm
Write-up	https://dbourdeau.github.io/cyphersolver/armstrong.html

3. Existing keys found outside the collection

A **key** already existed but had not been connected with the document; it was found in another archive, volume, database record or publication and matched to the ciphertext.

35. Dr Nicholas Wotton (Soissons) to Queen Mary I 13 June 1554

To Queen Mary I

“... the Constable comenid with him how the ysle of Wight might be taken ... and how it might be kept ... He the French king hath encreasidde the pensions of dyvers of the rebels ...”

Dr Nicholas Wotton, English ambassador in France, Soissons, June 13, 1554

The key (homophonic sign alphabet, nulls, ~140-entry nomenclator) was matched to the letter and corrected over eight passes: 75% of 2,801 signs firm, 89% with tentative values. Carew and the rebels at Compiègne with the French king and the Constable.

Method	external key matched · partial
Primary source	British Library, Harley MS 1582 ff. 8-10
Secondary source	DECODE R8499 ("N. Wotttoy ? to unknown recipient")
Key or plaintext from	A key filed in another archive and series, not tied to the letter: TNA SP 106/2 f. 162, "France 1554" (DECODE R354)
Write-up	https://dbourdeau.github.io/cyphersolver/harley1582r8499.html

36. A correspondent at Randan to Louis de Gonzague, duc de Nevers 27 Mar - 4 Apr 1587

To Louis de Gonzague, duc de Nevers

“45 [le duc d'Espéron] a esté en tres grande alarme, passant pres Lyon et ce pais, et sans aucun subject, car je vous puis asseurer qu'on n'a jamais songé de luy faire desplaisir ...”

Translation: “45 [the duc d'Épernon] was in very great alarm, passing near Lyon and this country, and without any cause, for I can assure you that no one ever thought of doing him harm ...”

An unnamed correspondent of the duc de Nevers, [Randan], April 4, 1587

The key book entry fits unchanged once the writer's glyph forms were learnt from a glossed sibling letter (BnF fr. 3413 f. 102). 26 of 27 cipher units read: Épernon passing near Lyon; the first letter is dated in cipher "De Randan", not Rome.

Method	external key matched · complete
Primary source	BnF, français 3975 ff. 28, 30 · Gallica
Secondary source	DECODE R4285 , R4286 ("Unknown sender (Rome)")
Key or plaintext from	Key no. 11 of the Nevers key book, BnF français 3995 f. 23, not tied to these letters by DECODE, the BnF or Tomokiyo
Write-up	https://dbourdeau.github.io/cyphersolver/nevers1587.html

37. Robert Reade (Paris) to his cousin [Thomas Windebank?] 19 Apr 1641

To his cousin [Thomas Windebank?]

“... I am of opinion they have not enough against him to confiscate his fortune, and they would be [willing] to take the advantage of his refusall to return ...”

Robert Reade, nephew and secretary of Sir Francis Windebank, Charles I's fugitive Secretary of State, Paris, April 19, 1641

The Windebank key was found to read the letter; 11 values were rebuilt here. 94% read: Windebank's return under a summons, St Ravy and Mr Forster. The writer is identified from signature and endorsement as Windebank's nephew and secretary.

Method	external key matched · partial
Primary source	British Library, Harley MS 7001 ff. 148-149
Secondary source	DECODE R7766
Key or plaintext from	The Deciphering Branch reconstruction of the Windebank family cipher, in another volume, BL Add MS 32256 f. 4 (DECODE R9115)
Write-up	https://dbourdeau.github.io/cyphersolver/harley7001.html

38. Charles I to Prince Rupert Oxford, 29 Apr 1645

To Prince Rupert, his nephew

“... I desire you to [march] speedily for my assistance, that in case I can not bring my train of artillery to you ...”

Charles I, King of England, Oxford, April 29, 1645

Lasry's key reads the holograph unchanged: the King cannot convoy his artillery train and asks Rupert to march and fetch it. Groups beyond the key read from context; about fifteen code groups open.

Method	external key matched · complete
Primary source	British Library, Add MS 18983 f. 14
Secondary source	DECODE R4921 (Non-decryptd; no decipherment in print)
Key or plaintext from	George Lasry's reconstruction of the King-Queen cipher from other letters, TNA SP 106/10 (DECODE R929)
Write-up	https://dbourdeau.github.io/cyphersolver/rupert1645.html

39. Bordeaux to Brienne (Thurloe intercept) London, 30 May 1653

To the Comte de Brienne, Secretary of State for Foreign Affairs

“Ce n’est pas que si nos affaires estoient plus florissantes nous n’eussions sujet de tesmoi[g]ner quelque mescontentement de la reception qui a esté faite aux deputez de Bordeaux. Ils ont esté entendus par deux commissaires du Conseil, et l’audiance que j’avois demandée au General ne me fut donnée qu’après la leur ...”

Translation: “*Not that, were our affairs more flourishing, we would lack cause to show some displeasure at the reception given to the deputies of Bordeaux. They were heard by two commissioners of the Council, and the audience I had asked of the General [Cromwell] was not granted me until after theirs ...*”

Antoine de Bordeaux-Neufville, French envoy in England, London, May 30, 1653

After a ciphertext-only solver failed on a faulty transcription, the key sheet was found and fitted: alphabetical syllabaries in three marked series. With 21 transcription corrections the despatch reads: the Council hears the Bordeaux (Ormée) deputies before Cromwell receives the envoy.

Method	external key matched · complete
Primary source	British Library, Add MS 4200 ff. 88-90
Secondary source	DECODE R8390 (duplicates R8391 , R8393); Tomokiyo, cryptiana (louisxivo.htm) · louisxivo.htm
Key or plaintext from	The English Deciphering Branch's key sheet "To Mr. Bordeaux French M.r at London 1653" in a separate key volume, BL Add MS 32263 f. 1 (DECODE R7537)
Write-up	https://dbourdeau.github.io/cyphersolver/bordeaux1653.html

40. Johan Isaac de Swart (St Petersburg) to Pieter van Bleiswijk 8 Mar 1782

To Pieter van Bleiswijk, Grand Pensionary of Holland

“... of nu den Keizer in dezen met haar oprecht handelt ... om haar te amuseeren ...”

Translation: “*... whether the Emperor is now dealing honestly with her [Catherine II] in this ... to string her along ...*”

Johan Isaac de Swart, Dutch resident in Russia, St Petersburg, March 8, 1782

The 1765 codebook, parsed to 5,961 entries, was found to fit and read 84% as transcribed; 34 values from context and 41 entries read from the key scans raised it to 96.7% of 1,719 groups. The French and Spanish answers to the Austro-Russian mediation, Catherine II's Greek project.

Method	external key matched · complete
Primary source	Nationaal Archief, 3.01.25 (Van Bleiswijk) inv. 610
Secondary source	DECODE R1036 (codebook "could not be identified")
Key or plaintext from	Croiset's 1765 codebook for the Russia legation, a separate key record (DECODE R1038)
Write-up	https://dbourdeau.github.io/cyphersolver/deswart1782.html

41. Prince Frederick of Orange (London) to the Hereditary Prince 7 May 1795

To the Hereditary Prince of Orange, his brother

“... men heeft de ratificatie ... gepubliceert, of schoon de ratificatie zelf nog niet gekoomen was ... maar blijft persisteren [...] het tegenwoordig gouvernement niet te erkennen, dan zoo zij geen Robespierismus durven te introduceeren ...”

Translation: “... *they have published the ratification ... although the ratification itself had not yet arrived ... but persists [...] in not recognising the present government, unless they dare not bring in Robespierism ...*”

Prince Frederick of Orange, younger son of the exiled Stadtholder William V,
London, May 7, 1795

The R1892 key reads this letter unchanged; the clear Dutch under page 1 confirms it and fixes twelve word signs. Pages 3-4: scattered émigré troops, the Basel ratification, ambassadors to Paris. About twenty word signs open.

Method	external key matched · partial
Primary source	Koninklijk Huisarchief, The Hague, Koning Willem I inv. XVIII-3
Secondary source	DECODE R2242 (non-decrypt)
Key or plaintext from	The 6x6 digit-square key recovered ciphertext-only in this project for a letter in another fonds, Prins Willem V inv. 339 (DECODE R1892)
Write-up	https://dbourdeau.github.io/cyphersolver/r2242.html

4. Easier cases: key or plaintext found elsewhere in the same collection

The matching key or plaintext came from the same collection (another volume or fonds, no link recorded before), but had not been previously associated with the unsolved ciphertext.

42. Lope de Soria to Charles V: the key-B letters Genoa, Jul-Aug 1523

To Emperor Charles V

“Andrea Doria y el principe de Salerno esta[n] muy descontentos del rey ... si V. Mag. [quisiere] aceptar en su servicio al dicho Andrea Doria con sus [qua]tro galeras, que el lo servira de muy buena voluntad ...”

Translation: “*Andrea Doria and the Prince of Salerno are very discontented with the King [of France] ... if Your Majesty [should wish] to take the said Andrea Doria into his service with his [four] galleys, he will serve him very willingly ...*”

Lope de Soria, Charles V's ambassador, Genoa, August 13, 1523

Key B was rebuilt by aligning R9844 with its decipherment; the key-B letters read in substance (Siena, Beaurain, money for Swiss to descend on France, Andrea Doria). The earlier key-A letters were broken ciphertext-only and are listed in section 1.

Method	key from external plaintext · partial · split entry: key B (from late July) only
Primary source	Real Academia de la Historia, Colección Salazar y Castro A-28 (9/28)
Secondary source	DECODE R9491 - R9498
Key or plaintext from	The court's decipherment of another Soria letter (30 Dec 1523) in a different volume of the same collection, Salazar A-30 (DECODE R9844)
Why borderline	same collection and correspondent, different volume
Write-up	https://dbourdeau.github.io/cyphersolver/soria1523.html

43. Duke of Sessa to Charles V Feb-Apr 1523

To Emperor Charles V

“... Su Santidad por esta bula es declarado contra Francia ...”

Translation: “... *His Holiness, by this bull, stands declared against France ...*”

Luis Fernández de Córdoba, Duke of Sessa, Charles V's ambassador,
Rome, April 26, 1523

The key was recognised by recurring code groups and applied unchanged. Mostly already in print; the two uncalendared despatches (R9660, R9665) read in part.

Method	external key matched · partial
Primary source	Real Academia de la Historia, Colección Salazar y Castro A-27 (9/27)
Secondary source	DECODE R9660-R9666
Key or plaintext from	The Sessa key this project rebuilt from the court's decipherments in another volume of the same collection (Salazar A-31); four of the letters are calendared in CSP Spain ii (1866) · BHQ
Why borderline	same collection and correspondent, different volume
Write-up	https://dbourdeau.github.io/cyphersolver/sessa1523.html

44. András Bay to Ferenc Rákóczi II Iași, 8 Mar 1706

To Ferenc Rákóczi II, Prince of Transylvania

“... tudván az barát az Portának szemének és fülének lenni, még nem mondotta ...
lelkemre kötötte, hogy ki ne adjam őtet. A svéd Vilnához szállott ...”

Translation: “... *knowing the friar to be the eyes and ears of the Porte, he has not yet told it ... he charged me on my soul not to give him away. The Swede has taken up quarters at Vilna ...*”

András Bay, Rákóczi's envoy at the court of Moldavia, Iași, March 8, 1706

The key reads the letter unchanged (97.6% as sense) after seven transcription corrections: Muscovite and Polish envoys at the Moldavian court, the Tsar suing Charles XII for peace.

Method	external key matched · complete
Primary source	MNL OL, G15 Caps. D. Fasc. 81. 5
Secondary source	DECODE R478 (non-decrypt)
Key or plaintext from	Rákóczi's syllabic key, G15 Caps. C. Fasc. 43. 56 (DECODE R609), picked by scoring all 94 G15 key transcriptions against the letter
Why borderline	same fonds (G15), different caps; no link recorded before
Write-up	https://dbourdeau.github.io/cyphersolver/bay1706.html

45. Philippe Groffey (?) to Ferenc Rákóczi II (alias Pompeo Csoni) 15 Oct 1707

To Ferenc Rákóczi II (addressed as “Monsieur Pompeo Csoni”)

“... prendre les intérêts de la Pologne autant à cœur ... se sacrifier ... pour affranchir la liberté ... opprimée par les Suédois en faveur du roi Stanislas ...”

Translation: “... *to take the interests of Poland as much to heart ... to sacrifice himself ... to set free the liberty ... oppressed by the Swedes in favour of King Stanislas ...*”

Philippe Groffey (?), Rákóczi's agent, October 15, 1707

The key identifies 1,020 of 1,108 groups (92.1%), validated on two interlined siblings; the catalogue's direction is reversed (Rákóczi is the recipient). Ráday's letters to Rehnskiöld, Polish liberty oppressed in favour of King Stanisław.

Method	external key matched · partial
Primary source	MNL OL, G15 Caps. C. Fasc. 39 pp. 277-279
Secondary source	DECODE R902 (non-decrypt)

Key or plaintext from	The key "De Monsieur de Bonac et Graffei", G15 Caps. C. Fasc. 44/o8 (DECODE R639), found by scoring the G15 keys
Why borderline	same fonds (G15), different fascicle; no link recorded before
Write-up	https://dbourdeau.github.io/cyphersolver/rakoczi1707.html

46. Letters to Ferenc Rákóczi II (Poland and Danzig) Oct-Nov 1707

To Ferenc Rákóczi II

“... fait à Madame la Palatine de Belz ... une copie authentique de cette vente au sieur Kray ... autant qu’il a esté possible par mon credit et par mes avances ...”

Translation: “... made to Madame the Palatine of Belz ... an authentic copy of this sale to the sieur Kray ... as far as has been possible through my credit and my advances ...”

An unsigned correspondent, Danzig, October 28, 1707

R912 and R922 keyed at 97.4% of 2,352 groups: Stanisław, Charles XII, the Tsar, Rehnskiöld. The other two letters of the group were read from their own clear copy and interlinear decipherment and are not counted here.

Method	external key matched · complete
Primary source	MNL OL, G15 Caps. C. Fasc. 39
Secondary source	DECODE R912 , R922 (catalogued as Rákóczi to unknown recipient)
Key or plaintext from	The same Bonac-Groffey key, G15 Caps. C. Fasc. 44/o8 (DECODE R639)
Why borderline	same fonds (G15), different fascicle
Write-up	https://dbourdeau.github.io/cyphersolver/rakoczi1704.html

47. London to Johann Georg von Ilten Feb-Apr 1743

To Johann Georg von Ilten, Hanoverian lieutenant-general in the Low Countries

“Cependant on souhaite que les troupes consomment ... ce qu’il y a de fourrage dans ces quartiers, afin qu’il n’en reste rien si l’envie prenoit aux François de vouloir prendre quartier après vous dans ce pays là, de sorte qu’ils n’y trouveroient point ni fourrage ni rien de ce qui seroit necessaire pour la subsistance des troupes.”

Translation: “Nevertheless it is wished that the troops should consume ... whatever forage there is in these quarters, so that none is left should the French take a fancy to quarter in that country after you, so that they would find there neither forage nor anything needed for the subsistence of the troops.”

An unsigned writer, on George II's orders, London, February 15/26, 1743

The code table was matched to the letters through the worksheet values of one of them and reads both: 220 of 223 groups. The King approves the forage instructions so the French find none; an extract of Lord Stair's instructions.

Method	external key matched · complete
Primary source	GWLB Hannover, Ms XXIII 1234:29,2 pp. 73-75, 119-121 · GWLB
Secondary source	Catalogue search of the Ilten papers
Key or plaintext from	The "Chiffre-Tabelle" in another volume of the same papers, Ms XXIII 1234:31,1 pp. 386-391 (1745/46)
Why borderline	same collection, different volume
Write-up	https://dbourdeau.github.io/cyphersolver/ilten1743.html